

Finding a Three-Way Collision with Bounded Memory

The Oblivious Sequential Time–Memory Curve

Status. Statement: AI-written, not yet formalized.

Category. Symmetric-Key Cryptography.

Keywords. 3-collisions; multi-collision finding; branching programs; time–space tradeoffs; oblivious algorithms; streaming switching lemma; bounded-memory cryptanalysis.

Conjecture (informal). *Fix a schedule of queries to a random function in advance – an oblivious algorithm – and allow it S bits of working memory. Finding three inputs with a common output should cost about $N/\sqrt{S}$ queries once S exceeds polylogarithmic size, until this curve meets the unconditional threshold of $N^{2/3}$ queries at $S \approx N^{2/3}$, beyond which memory buys nothing further. Both endpoints are already known – a proved theorem at polylogarithmic S , and the query threshold at any memory – and a matching algorithm (build a table of two-way collisions, then hunt a third match to one of them) achieves the curve throughout. The conjecture is that no oblivious algorithm can beat it.*

Setting. Throughout, $N \geq 2$ and $f: [N] \rightarrow [N]$ is a uniformly random function, available only through an oracle answering $x \mapsto f(x)$. A k -collision is a set of k distinct points of $[N]$ with a common image under f ; a 3-collision is the case $k = 3$. Asymptotic notation $\tilde{O}, \tilde{\Omega}, \tilde{\Theta}$ suppresses factors polylogarithmic in N . Success probabilities are over the choice of f and any coins of the algorithm; randomisation need not be allowed for separately, since the input is already random.

What is known. Three facts fix the landscape.

(F1) *The query threshold is $N^{2/3}$. Among q queried points the expected number of 3-collisions is $\binom{q}{3}N^{-2}$, so $q = \Theta(N^{2/3})$ queries are necessary and sufficient for constant success prob-*

ability (Suzuki, Tonien, Kurosawa and Toyota, ICISC 2006). Quantitatively, an algorithm making T queries and then naming three points succeeds with probability $O((T^3 + 1)N^{-2})$. This holds for *any* algorithm, at any memory bound, and the conjecture below does not contradict it.

- (F2) *Without a table*, $\Theta(N)$ queries suffice. Find a 2-collision $\{u, v\}$ by cycle-finding, retain u , v and $y = f(u)$, then sample fresh points until one maps to y . This uses $O(\log N)$ bits of memory. Between **F1** and **F2** lies a gap of $N^{1/3}$, and the conjecture below asserts that memory is what closes it.
- (F3) *At $k = 2$ memory is free*. Pollard's rho method with Brent's cycle detection finds a 2-collision in $\Theta(\sqrt{N})$ queries using $O(\log N)$ bits, and that count is optimal even with unbounded memory. Consequently the conjecture below is *false* at $k = 2$, and no proof of it can be indifferent to the collision order.

Model. A *branching program* of length T and width W – the model of Borodin and Cook (SICOMP 1982), specialised to a random-function oracle – is a DAG on levels $0, \dots, T$ with at most W nodes per level, in which every node below level T carries a label $x \in [N]$ and has one outgoing edge for each $y \in [N]$, and every node at level T carries a label in $[N]^3$. It is run by starting at the source, querying the label of the current node, following the edge labelled by the answer, and reading the label of the sink reached; it *succeeds* if that label is a 3-collision. Its *space* is $S = \log_2 W$, counted in bits. The state may encode anything, so one cannot say what a program *knows*. It is *oblivious* if all nodes at a level carry the same label, so that the query schedule $w_1, \dots, w_T$ is fixed in advance, and *repetition-free* if no point labels two levels.

The proved base. The only lower bound known is for oblivious programs, where the multi-pass streaming switching lemma of Dinur (EUROCRYPT 2020) applies to the answer stream. For *adaptive* programs nothing memory-sensitive is known at $k = 3$: the only available lower bound is the query bound **F1**, which mentions no memory at all.

Theorem BSo (oblivious programs). *Every repetition-free oblivious branching program succeeding with probability at least $\frac{1}{2}$ satisfies $T \cdot (S + \lceil \log_2 N \rceil) = \tilde{\Omega}(N)$; in particular $T = \tilde{\Omega}(N)$ at width $N^{O(1)}$.*

The conjecture. It sharpens this, and is tight against an algorithm.

Conjecture BSo⁺ (the oblivious sequential curve at $k = 3$). *Every repetition-free oblivious branching program succeeding with probability at least $\frac{1}{2}$ satisfies*

$$T = \tilde{\Omega}(\max\{N^{2/3}, N/\sqrt{S}\}),$$

$$\text{equivalently} \quad T \cdot S^{1/2} = \tilde{\Omega}(N) \text{ for } S \leq N^{2/3}.$$

The first clause is **F1** and carries no content; all of it is in $T \cdot S^{1/2}$. The bound is tight: an oblivious program can accumulate $P \leq S$ collision pairs at cost $\tilde{\Theta}(PN/S)$ and then hunt a third preimage of one of the P recorded values in $\tilde{\Theta}(N/P)$ further queries, both phases non-adaptively; balancing at $P = \sqrt{S}$ gives $T = \tilde{\Theta}(N/\sqrt{S})$. So a factor $\sqrt{S}$ separates **BSo** from the conjecture inside its own class. The two clauses cross at $S = N^{2/3}$, where the curve meets the query threshold and the problem closes.